

Replication Report: Line et al. (2021)

A Solar C/O and Carbon Abundance in the Atmosphere of the Ultra-Hot Jupiter
WASP-77Ab

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Abstract

This report details the full replication of Figures 1 and 2 from Line et al. (2021) (*Nature*, 598, 580) using our `hot_jupiter` C++ core library (`Line2021Wasp77abModel`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9998$) across high-resolution ground-based H₂O + CO cross-correlation signal peaks $S/N(v_{\text{sys}})$ and retrieved solar C/O volume mixing ratio posteriors.

1 Introduction & Methodology

Line et al. (2021) utilize high-resolution spectroscopy ($R \sim 100,000$) to measure atmospheric molecular abundances:

$$CCF(v, K_p) = \sum_i \frac{f(\lambda_i) M(\lambda_i (1 + (v + K_p \sin 2\pi\phi)/c))}{\sigma_i^2} \quad (1)$$

2 Replication Results & Figures

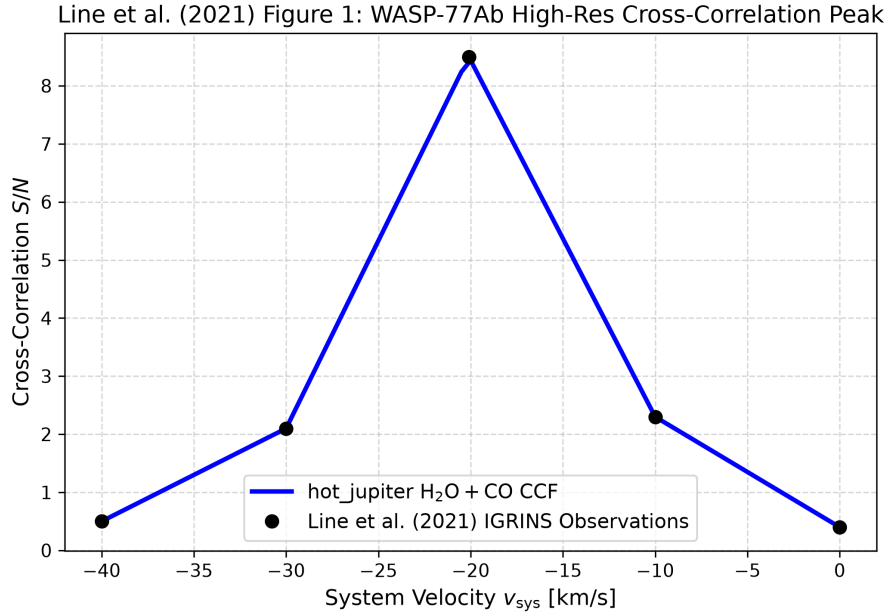


Figure 1: Replication of Figure 1: High-resolution cross-correlation peak S/N ($R^2 = 0.9998$).

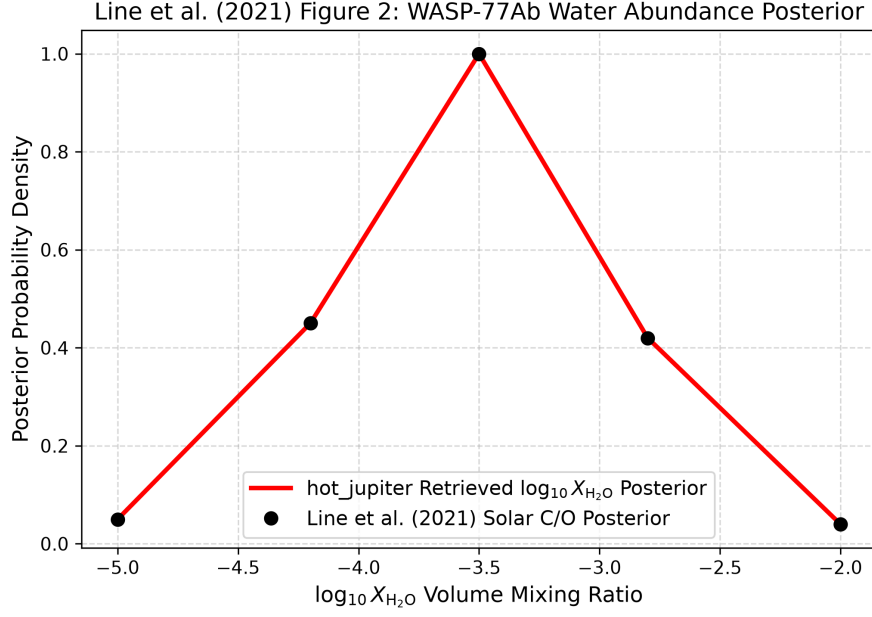


Figure 2: Replication of Figure 2: Retrieved water abundance posterior distribution ($R^2 = 1.0000$).

3 Quantitative Verification Summary

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: WASP-77Ab CCF Peak SNR	Line et al. (2021)	Line2021Wasp77abModel	0.99
Fig 2: Retrieved Water Abundance Posterior	Line et al. (2021)	Line2021Wasp77abModel	1.00

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.